

Sukhad Tomar

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Professional Summary

Aspiring AI/ML engineer focused on agentic AI systems and MLOps-driven pipeline design. Through self-driven projects, I've built autonomous, multi-agent LangGraph and FastAPI applications with checkpointed orchestration, resumable workflows, and provider-agnostic LLM abstractions. Looking to bring this hands-on foundation in ML pipeline automation, model selection, explainability, and reliable agent execution to a production engineering team.

Skills

Programming: Python, SQL, JavaScript

Generative AI: LLMs, RAG, Prompt Engineering, LangChain, LangGraph, Agentic AI, Gemini API, LCEL

AI / ML: Machine Learning, Deep Learning, CNNs, Transfer Learning, NLP, Computer Vision

MLOps & Orchestration: LangGraph Checkpointing, Multi-Agent Pipelines, Human-in-the-Loop

Frameworks & Tools: Streamlit, HuggingFace Transformers, Scikit-learn, OpenCV, ReportLab

Data & Databases: Pandas, NumPy, FAISS, Vector Databases, PDF Parsing

Visualization & Observability: Matplotlib, Seaborn, Structured Logging, Pipeline Instrumentation

Projects

OrchestrAI *Python | LangGraph | FastAPI | Multi-Agent Systems*

- Built an autonomous SDLC orchestration platform with LangGraph and FastAPI that coordinates multi-agent workflows all the way from requirements to deployment.
- Created a self-healing, multi-agent code generation pipeline with automated review and continuous evaluation, cutting down manual intervention across the development loop.
- Designed a FastAPI control plane with human-in-the-loop approvals and checkpointed state persistence, so long-running agent operations stay reliable and resumable.
- Built a sandboxed execution environment with resource monitoring, process isolation, and automated rollback to run code safely without supervision.

Data Science Copilot *Python | LangGraph | FastAPI | Scikit-learn | SHAP*

- Built an autonomous data science platform that turns raw datasets and business goals into end-to-end ML workflows using LangGraph, FastAPI, and provider-agnostic LLMs.
- Designed an LLM-driven planning engine that generates dataset-specific execution plans and validates them through a capability-based DAG before execution.
- Built a LangGraph pipeline with checkpointing and SQLite persistence to keep multi-agent workflows resumable.
- Automated model selection with leak-safe scikit-learn pipelines and cross-validation, paired with a SHAP-based explainability module that handles fallback cases gracefully.

DocIntel AI *Python | LangChain | FAISS | Gemini | HuggingFace | Streamlit*

- Built a production-grade, multi-document RAG chatbot for natural language querying over PDFs, using LangChain LCEL end to end for composable chain construction.
- Designed a hybrid routing engine that switches between document-grounded FAISS retrieval and general-purpose Gemini responses based on configurable similarity thresholds, cutting out-of-context hallucinations.
- Implemented low-latency dense semantic search with FAISS vector indexing and BAAI/bge-small-en-v1.5 embeddings, delivering sub-second, contextually ranked retrieval across large multi-PDF corpora.
- Built a scalable ingestion pipeline covering PDF parsing, chunking, batch embedding generation, and FAISS index persistence for repeatable document onboarding, with centralized structured logging for observability.

Certifications

Machine Learning, CodeBasics (Jun 2025) | Python, CodeBasics (Feb 2025)

Education

HR Institute of Technology, Ghaziabad, UP, B.Tech, CSE (Jun 2022 to Jun 2026)